

BYEONGHWA JEONG

Ph.D. AT GEOGRAPHY (GEOSPATIAL DATA SCIENCES)

R&D Associate, Geospatial Science and Human Security Division, Oak Ridge National Laboratory,

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EDUCATION

University College London

London, UK

Ph.D. in Geography, Department of Geography

Sep 2019-Jun 2023

- Thesis: "An Integrated Deep Learning Model using Retail, Residential and Footfall Changes data to Predict Retail Gentrification"
- Supervisor: Prof Paul Longley and Prof James Cheshire
- Developed a bespoke AI model (Autoencoder-LSTM) to predict gentrified high streets across England by analysing longitudinal changes in residential, retail, and footfall variables.
- Estimated the long-term impact of the COVID-19 pandemic on retail gentrification using Multivariate Multiple Regression and a Likelihood Ratio Test, finding that its effects were insignificant.
- Supervisors: Professor Paul Longley and Professor James Cheshire

The University of Edinburgh

Edinburgh, UK

Master of Science in Geographical Information Sciences, School of Geosciences

Sep 2017-Aug 2018

- Dissertation: Anticipation of Commercial Gentrification through Time and Space Seoul, South Korea (Distinction).
- Predicted retail gentrification in Seoul using a Self-Organising Map (SOM) and Cellular Automata Markov model with land price, retailer, and socioeconomic data.

Kyungpook National University

Daegu, South Korea

Bachelor of Arts in Geography

Mar 2008-Aug 2015

RESEARCH AND PROFESSIONAL EXPERIENCE

Oak Ridge National Laboratory

Knoxville, USA

R&D Associate

Feb 2025-present

- Participating LandScan Global and Mosaic projects that generate 1km and 100m global gridded population data
- Integrating satellite-based damage assessments with population models to generate near real-time estimates of populations affected by disasters and conflicts
- Developing a global habitable probability mask using AI to improve the accuracy of gridded population data
- Building an AI model to predict building-level wealth indices from multi-modal data (satellite images, building features and building footprint data) to provide critical insights for urban inequality and policy
- Applying multiple instance learning for the survey-based data to produce more reliable small-area estimates of household wealth distribution.

The University of Toronto

Toronto, Canada

Post-Doctoral Research Fellow

Jul 2023-Jan 2025

- Participated in a downtown recovery project that measure the degree of recovery of downtown areas from the COVID-19 pandemic.
- Estimated the uncertainty of sample footfall data.
- Analysed the recovery rates of individual buildings located in downtown areas and investigated the relationship between building characteristics and recovery rates.
- Imputed missing footfall data using deep learning-based imputation models.
- Developed new downtown boundaries using unsupervised image segmentation models and open-source spatial big data.
- Explored the influence of socioeconomic variables on downtown recovery rates through explainable AI.

Daejeon Sejong Research Institute

Daejeon, South Korea

External Collaborative Researcher

Oct 2024-Dec 2024

- Participated in a research project "Big Data-Driven Strategies for Enhancing the 'Science City Daejeon' Brand: Focusing on the 2024 Korea Science Festival and Daejeon Science Festival".
- Crawled web contents about Korea Science Festival and Daejeon Science Festival between 2019 and 2024
- Conducted preprocessing of the collected text data and performed text mining through TF-IDF, Latent Dirichlet Allocation and network analysis.
- Explored changes in people's perceptions of science festivals.

Daejeon Sejong Research Institute

External Collaborative Researcher

Daejeon, South Korea

Aug 2023-Oct 2023

- Participated in a research project "Discover 'Regional Things' using Big Data Analysis and Civic Collaboration".
- Crawled text data from the diverse internet sites.
- Conducted preprocessing of the collected text data and performed text mining through TF-IDF, Latent Dirichlet Allocation and network analysis.

Daejeon Sejong Research Institute

External Technical Consultant

Daejeon, South Korea

Jul 2022-Sep 2022

- Participated in a research project "An Analysis of the Concept of Hip and Hot Place in Daejeon Using Social Media Data and Text Mining".
- Advised how to crawl text data on the internet using Python and various packages
- Technical consulting to conduct preprocessing for the crawled text data for the text mining
- Consulted how to perform text mining, particularly Latent Dirichlet Allocation and network analysis for text data
- Overall advice to conduct network analysis for text analyses

Korea Environment Institute

Part-time modeler

Sejong, South Korea

Aug 2021-Oct 2021

- Built a spatial interpolation model based on deep learning in order to more accurately predict unobserved areas by Orbiting Carbon Observatory-2 (OCO-2) satellite.
- The model interpolated unobserved areas by combining CO2 modeling (GEOS-Chem) and OCO-2 data.
- This model achieved superior results than traditional interpolation methods.

Korea Research Institute for Human Settlements

Assistant Research Fellow

Sejong, South Korea

Sep 2018-Sep 2019

- Participated in a study on the usage of national territory monitoring indicators for policymakers. Participated in a study on the analysis and application of the database of national territory monitoring indicators.
- Built accessibility indicators to daily infrastructures (schools, parks, etc.) based on a grid-based map covering South Korea.
- Created an old building proportion data for the grid-based map using a building dataset recording the trajectory of all buildings in South Korea

Kyungpook National University

Student Research Assistant

Daegu, South Korea

Nov 2015-Jun 2016

- Took part in "Evaluating sensor signature changing trends of GHG satellite caused by UN forestry carbon project"
- Based on the Orbiting Carbon Observatory-2 satellite data, explored the distribution of carbon in urban, in the case of Seoul.
- Learned the basic techniques of python to preprocess satellite data, also basic geostatistic theories and applications.

PUBLICATIONS**Peer-reviewed journal publications (published)**

- [P10] Bokyeong Lee; **Byeonghwa Jeong (corr. author)**, Development of Grid-Based Population Projection Method: A Modified Cohort Component Approach Applied to South Korea. *Population, Space and Place* 32, 1 (2025). <https://doi.org/10.1002/psp.70161>
- [P9] Xiaoxin Zhou; **Byeonghwa Jeong (corr. author)**; Karen Chapple, A Large Language Model-based Chatbot System Framework for Urban Planners, *Expert Systems With Applications* 130307 (2025). <https://doi.org/10.1016/j.eswa.2025.130307>
- [P8] **Byeonghwa Jeong**; Bokyeong Lee. Popnet: computer vision based bespoke deep learning model for forecasting gridded population. *International Journal of Geographical Information* 0, 0, 1–20 (2025). <https://doi.org/10.1080/13658816.2025.2514792>
- [P7] **Byeonghwa Jeong**; Juhwan Song; Younghoon Kim. A first open retail boundary dataset for South Korea using open data and computer vision technique. *Scientific Data* 12, 622 (2025). <https://doi.org/10.1038/s41597-025-04958-1>
- [P6] **Byeonghwa Jeong**; Karen Chapple. Does Class Matter? Understanding Differential Pandemic Recovery via a Building Typology. *Environment and Planning B: Urban Analytics and City Science* 0, 0 (2025). <https://doi.org/10.1177/23998083251326128>
- [P5] Amir Forouhar; Karen Chapple; Jeff Allen; **Byeonghwa Jeong**; Julia Greenberg. Assessing Downtown Recovery Rates and Determinants in North American Cities After the COVID-19 Pandemic. *Urban Studies* 62, 6, 1209-1231 (2024). <https://doi.org/10.1177/00420980241270987>
- [P4] **Byeonghwa Jeong**; Jeff Allen; Karen Chapple. A new commercial boundary dataset for metropolitan areas in the USA and Canada, built from open data. *Scientific Data* 11, 422 (2024). <https://doi.org/10.1038/s41597-024-03275-3>
- [P3] **Byeonghwa Jeong**; Junwoo, Kim. Topic Modeling of the Transition of the Sense of Place in Urban Regeneration Area: A

Case Study of Daegu Bangcheon Market. *Journal of Daegu Gyeongbuk Studies* 19, 1 (2020):27-44

- [P2] **Byeonghwa Jeong**; Junwoo, Kim. Application and Evaluation of Self-Organising Map-based Spatial Clustering for Regional Policy Recommendations. *Journal of the Korean Geographical Society* 54, 3 (2019):387-404.
- [P1] **Byeonghwa Jeong**; Jung-Sup Um. Comparative evaluation of South versus North Goseong County in terms of forest carbon offset potential. *Spatial Information Research* 24, 1 (2016).

Peer-reviewed journal publications (*under review*)

- [U2] **Byeonghwa Jeong**; Junwoo Kim, Prediction of retail Gentrification Using a Data Science Approach; Case Study in Seoul, South Korea, *Environment and Planning B* (under review)
- [U1] Changwha Oh; Gaweon Shin; **Byeonghwa Jeong (corr. author)**. Exploring visitor-driven tourism themes: A bottom-up analysis using topic modelling of geotagged user-generated contents. *International Journal of Tourism Research* (under review).

Reports

- [R8] **Byeonghwa Jeong**; Viswadeep Lebakula; Marie Urban, PopGNN: Graph Neural Network-Based Flexible Future Population Forecasting Model, Oak Ridge National Laboratory (ORNL), Oak Ridge, TN (United States) (2026)
- [R7] **Byeonghwa Jeong**; Daniel Adams; Andrew Zimmer; Marie Urban, Satellite Embedding-Based Population Imputation for Areas with Missing Building Footprint Data: A Computer Vision-Based Approach, Oak Ridge National Laboratory (ORNL), Oak Ridge, TN (United States) (2026)
- [R6] Karen Chapple; Amir Forouhar; **Byeonghwa Jeong**; Stephanie Ortynsky, Assessing the Doom Loop, Lincoln institute of land policy (2025)
- [R5] Hyejin Joo; Sangheon Han; Eunjeong Jeong; **Byeonghwa Jeong**; Juhwan Song; Minyoung Noh, Big Data-Driven Strategies for Enhancing the 'Science City Daejeon' Brand: Focusing on the 2024 Korea Science Festival and Daejeon Science Festival, Daejeon Sejong Research Institute (2025)
- [R4] Hyejin Joo; Minyoung Noh; **Byeonghwa Jeong**, Basic Research for Citizen Participatory Urban Branding: Focusing on Human and Cultural Resources, Daejeon Sejong Research Institute (2024)
- [R3] Hyejin Joo; Sangheon Han; Seolmin Yoon; Jaeyeon Moon; **Byeonghwa Jeong**, Discover 'Regional Things' using Big Data Analysis and Civic Collaboration, Daejeon Sejong Research Institute (2023)
- [R2] Eunsun Lim; Youngju Lee; Bokyoung Lee; Yohan Jang; Jaesun Son; Kyungju Lee; Pyeongjik Lee; **Byeonghwa Jeong**; Hyungkwan Choi, Development and Utilization of Multiple Deprivation Index for Place-based Living Facilities Policy, Korea Research Institute for Human Settlements, (2021).
- [R1] Eunsun Lim; Youngju Lee; Bokyoung Lee; Pilsung Byun; Moonsoo Shin; **Byeonghwa Jeong**; Changhwa Oh; Gawon Shin; Kyungju Lee; Jongcheon Ahn; Hosang Lee; Jinho Park, National Territorial Monitoring Report in 2018, National Geographic Information Institute, (2019).

Work in preparation

- [P4] **Byeonghwa Jeong (corr. author)**; Joe Tuccillo; Building-Level Wealth Index Prediction Using DHS Survey Data and Machine Learning: A Flexible Multi-Scale Framework. in preparation for submission to *Computers, Environment and Urban Systems*.
- [P3] **Byeonghwa Jeong (corr. author)**; Andrew Zimmer; Daniel Adams; Satellite Embedding-Based Population Imputation for Areas with Missing Building Footprint Data: A Computer Vision-Based Approach. in preparation for submission to *ACM Knowledge Discovery and Data Mining (KDD)*.
- [P2] **Byeonghwa Jeong**; Viswadeep Lebakula; Brandyn Reynolds, Global level Livability Mask for Gridded Population Datasets. in preparation for submission to *Scientific Data*.
- [P1] Andrew Zimmer; Daniel Adams; **Byeonghwa Jeong**; Marie Urban. Framework for Rapid Updates to LandScan High-Resolution Gridded Population Datasets. in preparation for submission to *Scientific Reports*.

CONFERENCE PRESENTATION

- 2024 Young-Hoon Kim; **Byeonghwa Jeong**, Delineating commercial boundaries using an AI-based computer vision technique with open data in South Korea, AsiaCarto 2024
- 2024 **Byeonghwa Jeong**; Karen Chapple; Jeff Allen, Typology of Downtown Recovery at the Granular Scale, Geoinformatics2024

GRANTS

- [G3] A Study on the Evaluation and Characteristics of Coastal Area by Using Image Segmentation Model,
Korea Maritime Institute (KMI) \$2,885 (exchange rate 1\$ = 1,386.3 (Korea Won))
- Personal contribution: idea development, providing core data, writing and review
- [G2] Assessing the doom loop: Scenarios in four North American downtowns,
Lincoln Institute of Land Policy \$52,931 (exchange rate 1\$ = 1.37 CAD)
- Personal contribution: data curation, data collection, data preprocessing, explainable AI performing, writing proposal (some parts)
- [G1] MARSHALLING A ROBUST DOWNTOWN RECOVERY FOR OTTAWA,
Canadian Urban Institute \$109,514 (exchange rate 1\$ = 1.37 CAD)
- Personal contribution: data curation, data collection, data preprocessing, methodology selection, data analysis, visualization, writing proposal (some parts)

TEACHING AND SUPERVISION EXPERIENCE

Supervision

Undergraduate Students

- 2024 Christopher Kim (B.A in UC Berkeley, USA)
2024 Brandon Choi (B.A in UC Berkeley, USA)

Graduate Students

- 2024 Xiaoxin Zhou (M.S in Informatics, U of Toronto, Canada)
2024 Gaweon Shin (Ph.D in Public Administration, Yonsei University, South Korea)
2024 Changwha Oh (Ph.D in Geography, UT Knoxville, USA)

Guest Lecture

- 2025.3 The application of computer vision for delineating regional boundary, Center for territory monitoring, Korea Research Institute for Human Settlement (hosted by Dr Bokyeong Lee)
2024.8 The application of AI for social phenomenon prediction, Department of Geography Education, Korea National University of Education (hosted by Professor Younghoon Kim)
2023.6 Social phenomenon and spatial AI, Department of Geography, Kyungpook National University (hosted by Professor Jae Hyun Lee)

MANAGEMENT AND SERVICE

Conference and Event Management

- Member of the Organising Committee, International Conference on Geospatial Information Science, Spatially Enabled Society with AI and Digital Twin, 2019

Journal Reviewer

- International Journal of Geographical Information Science (IJGIS), Taylor & Francis, 2025
- Computers and Electrical Engineering, Elsevier, 2025
- Papers in Regional Science, Elsevier, 2026

HONOURS AND PRIZES

- [A3] Awarded best paper for "Topic Modeling of the Transition of the Sense of Place in Urban Regeneration Area: A Case Study of Daegu Bangcheon Market", Daegu Gyeongbuk Development Institute (Mar 2020)
- [A2] Awarded an outstanding paper award by presenting 'Comparative Evaluation of Pixel Value Signature Between UN-A/R CDM Project Area in Goseong, South Korea and Goseong, North Korea', Conference on Geo-Spatial Information, KOREA Spatial Information Society (May 2015)
- [A1] Awarded \$300 scholarship for the skilled ArcGIS person, June-Myeong Scholarship, GEO C&I Corp. (Mar 2013)

LANGUAGES

- Native in Korean
 - Reading: Native
 - Speaking: Native

- Writing: Native

TECHNICAL SKILLS

- GIS analysis: ArcGIS, QGIS (Skilled)
- Computation Language: Python (Skilled)
 - Data query and management: Pandas, Geopandas, Numpy, PySpark
 - Machine Learning and Deep Learning: PyTorch, Torchvision, scikit-learn, Gensim
 - Traditional Statistics: Diverse regressions (e.g. Multivariate Multiple Regression), Likelihood ratio test, Diverse correlation test (e.g. Pearson correlation coefficient), Feature engineering
 - Web-crawling: Selenium, BeautifulSoup, Request
 - Text-mining: TF-IDF, Word2vec, Latent Dirichlet Allocation
 - Large Language Models: Prompt engineering, fine-tuning
 - Network analysis: Networkx, UCINET
- Remote sensing: Google Earth Engine, Python-based raster analysis (Skilled)
- Cloud based server: Google Firebase and Colab (Skilled)
- Relational Database Management System: Mariadb (Have experienced)
- Statistic Software: R (Have experienced)